OpenMV® Multispectral Event Camera Module

Datasheet

SKU: OMV-MECM-PAG7936-V1

Description

The Multispectral Event Camera Module allows your OpenMV Cam to interface with the Prophesee GENX320 event vision sensor and the PAG7936 1 MP 120 FPS color global shutter sensor simultaneously. The PAG7936 exposure trigger is shared with the GENX320 event timestamp input, allowing synchronized event information with image capture. Inspired by the human retina, each sensor pixel responds independently and asynchronously to changes in illumination, enabling efficient, high-speed, and low-power vision sensing.

Target Areas

Event-based vision, high-speed motion detection, vibration analysis, autonomous systems, robotics, industrial inspection, low-power surveillance.

Key Features

- **Dual sensing** — Simultaneous GENX320 event camera + PAG7936 1 MP 120 FPS color global shutter.
- **Synchronized capture** — PAG7936 exposure trigger shared with GENX320 event timestamp for time-synchronized multispectral data.
- **375 Hz+ event rate** — Event histograms at 375 Hz+. No exposure time, no motion blur.
- **3 mW sensor power** — Ultra-low power event sensing starting at 3 mW streaming.
- **140 dB dynamic range** — Extended dynamic range — sees changes in both dark and bright zones simultaneously.
- **<5 lux low-light** — Senses into twilight light levels with 0.05 lux sensor cutoff.

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1 Specifications

1.1 Event Camera

Component	Details
Sensor	Prophesee GENX320 event vision sensor.
Event Rate	Event histograms at 375 Hz+.
Power	Starting at 3 mW streaming data.
Dynamic Range	Up to 140 dB.
Low-Light	Senses below 5 lux; sensor cutoff at 0.05 lux.

1.2 Color Camera

Component	Details
Sensor	PAG7936 1 MP color global shutter at 120 FPS.
Synchronization	Exposure trigger shared with GENX320 event timestamp input.

1.3 Compatibility

Component	Details
Supported Cameras	OpenMV N6 only.

2 General Info

Dimension	Value
Weight	5 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/multispectral-event-camera-module
Prophesee GENX320	https://www.prophesee.ai/event-based-sensor-genx320/
Prophesee Documentation	https://docs.prophesee.ai/stable/hw/partners/openmv.html
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.